Experimental Physics

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1 Measurement of Length

The simplest method to measure the length of a line segment is to use the metric scale. But there is a certain limit to the accuracy of the results:

(i) The marking lines on a scale have a finite thickness.

(ii) Estimates less than 0.5 mm cannot be measured by naked eyes correctly.

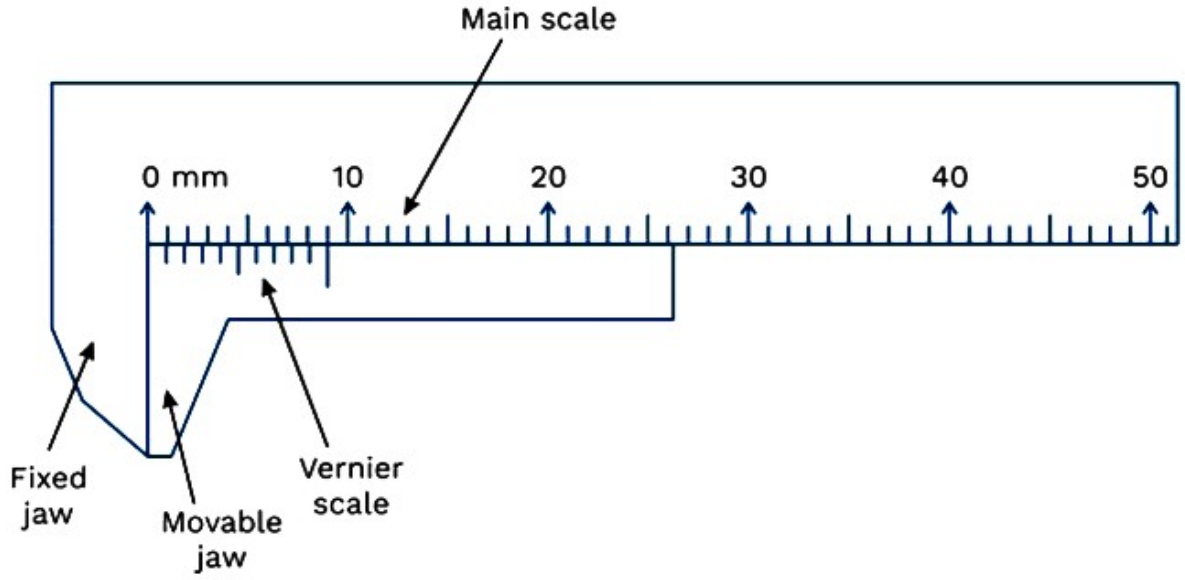
For greater accuracy, we use instruments like:

(a) Vernier calliper.

(b) Micro-meter scales (screw gauge).

Vernier Callipers :

Vernier callipers are used for measuring distance upto 0.1 mm accurately.



A vernier calliper is a combination of two scales. There is a main scale which is simply an mm scale. Each division on the main scale is of 1 mm length.

1 main scale division (1 MSD) = 1 mm.

The other scale is a movable scale known as vernier scale. This scale is movable and it can slide along the main scale with help of a suitable roller mechanism.

There are 10 divisions made on the vernier scale which exactly match with 9 divisions on the main scale.

10th line of vernier scale matches with 9th line of main scale.

10 vernier scale division (10 VSD) = 9 main scale division (9 MSD).

$\Rightarrow$ 1 vernier scale division (1 VSD) = $\frac{9}{10}$ main scale division $\left(\frac{9}{10}\text{MSD}\right)$

$\Rightarrow$ 1 VSD = 0.9 MSD = 0.9 mm ($\because$ 1 MSD = 1 mm)

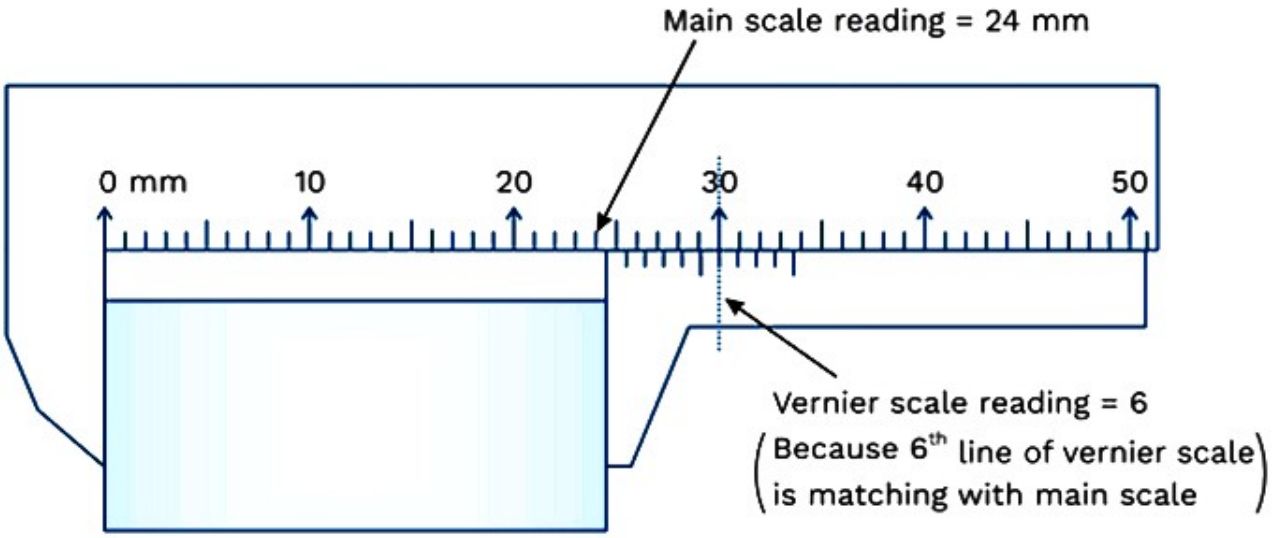
Least count of any instrument is minimum measurement which can be taken by that instrument.

Least count of vernier calliper = 1 MSD – 1 VSD

For above vernier calliper, least count = 1 mm – 0.9 mm = 0.1 mm

We fit the object between the jaws of the vernier calliper whose length is to be measured. After that the readings of the main scale and the vernier scale are taken.

Taking the reading of a Vernier Calliper:



Thickness of the object = 24. ---------

= 24 mm + 6 × (0.1 mm)

Formula says

Thickness of the object = [Main scale reading] + [Vernier scale reading] [Least count]

Vernier scale reading = Mark on vernier scale which exactly co-incides with some mark of main scale

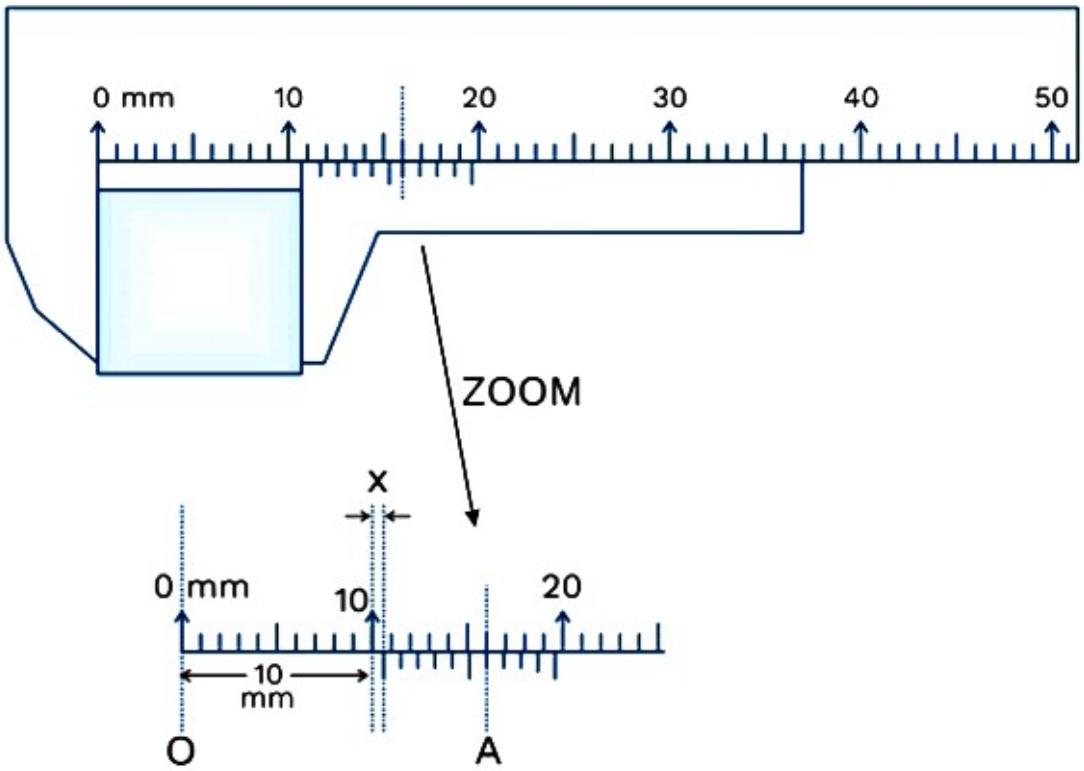
Least count = (1 MSD) – (1 VSD)

= 1 mm – 0.9 mm

= 0.1 mm

Thickness of the object = 24 + 0.6 = 24.6 mm.

Deriving the least count of Vernier Calliper



Consider a case when a block of dimension slightly more than 10 mm is fitted between the jaws of a vernier calliper as shown in the figure above.

Clearly, we can see that the length needed to be measured = 10 mm + x, where x = ?.

In the given figure, a vernier scale marking is exactly matching a main scale marking at point A.

Now, length OA must be same in two cases: when measured along the main scale and when measured along the vernier scale.

So, from the above figure,

10 mm + 6 (main scale division) = (10 mm + x) + 6 (vernier scale division)

$\Rightarrow$ 10 mm + x = 10 mm + 6 (main scale division − vernier scale division)

= 10 mm + 6 (1 mm − 0.9 mm)

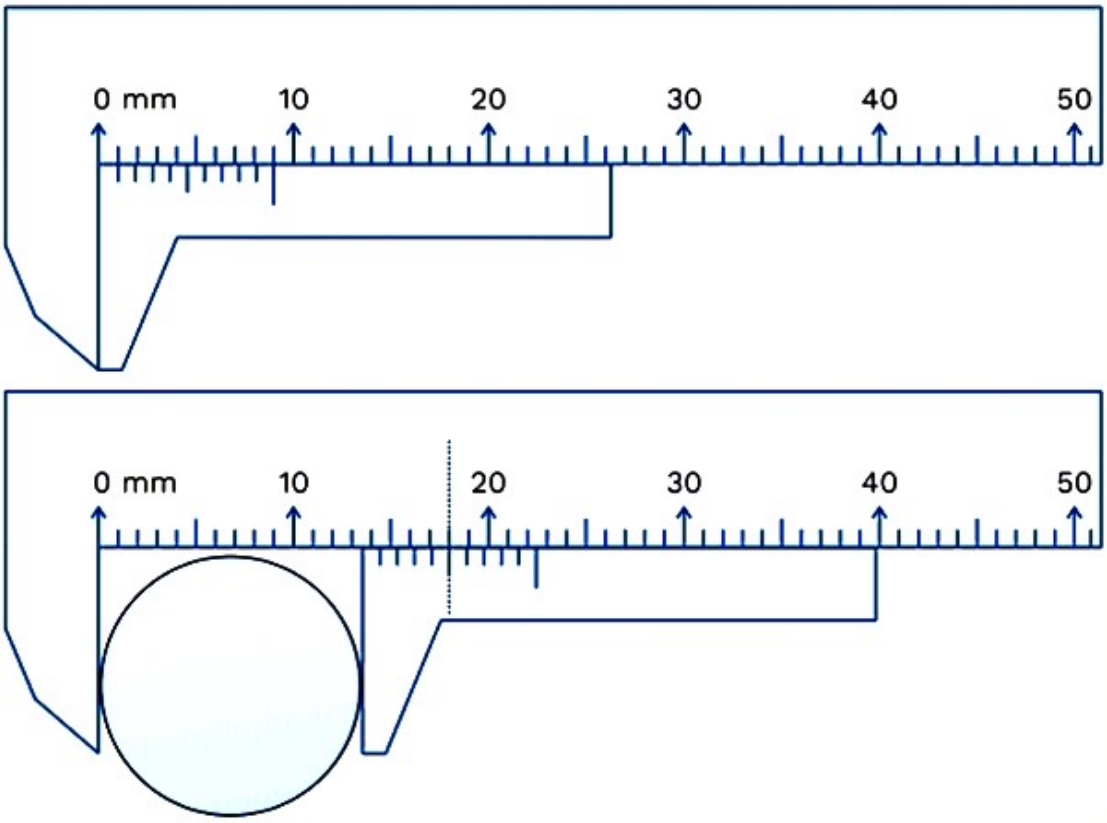
= 10 mm + 6 × (0.1 mm) = 10.6 mm

[Main scale reading] + [Vernier scale reading] × [Least count]

Thus, we can see that the least count is equal to the difference between 1 MSD (1 mm) and 1 VSD (0.9 mm).

$$\text{Thickness of object} = \text{Reading of vernier calliper} = \left[\text{Main scale reading}\right] + \left[\text{Vernier scale reading}\right] \times \left[\text{Least count}\right]$$

Q. **A vernier calliper is shown in two different situations. In first case there is no object between the jaws and in second case object is placed between the jaws. Find the diameter of object.**



Sol. As we know,

1 VSD = 0.9 mm

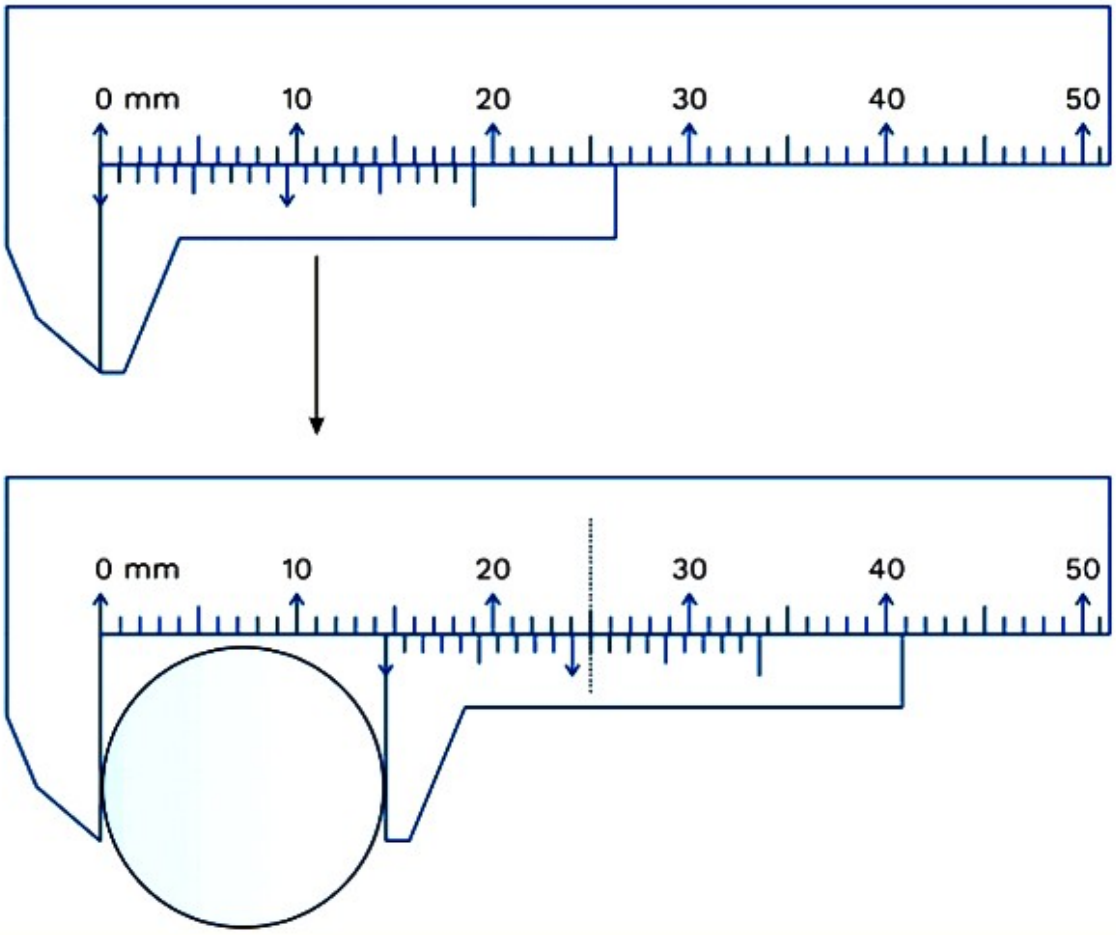
$\Rightarrow$ Least count = (1 MSD – 1 VSD)

= 1 mm – 0.9 mm = 0.1 mm

Diameter of the object = (Main scale reading) + (Vernier scale reading) (Least count)

So, diameter = 13 mm + (5) (0.1 mm) = 13.5 mm

Q. **A vernier calliper is shown in two different situations. In first case there is no object between the jaws and in second case object is placed between the jaws. Find the object's diameter.**



Sol. From the first figure, it can be seen that 20 divisions of vernier scale are coinciding with 19 division on the main scale.

So, 20 VSD = 19 mm

Then, 1 VSD $= \frac{19}{20}$ mm

Now, least count = (1 MSD – 1 VSD)

= 1 mm – 19/20 mm

= 0.05 mm

Diameter of the object = (main scale reading) + (vernier scale reading) (least count)

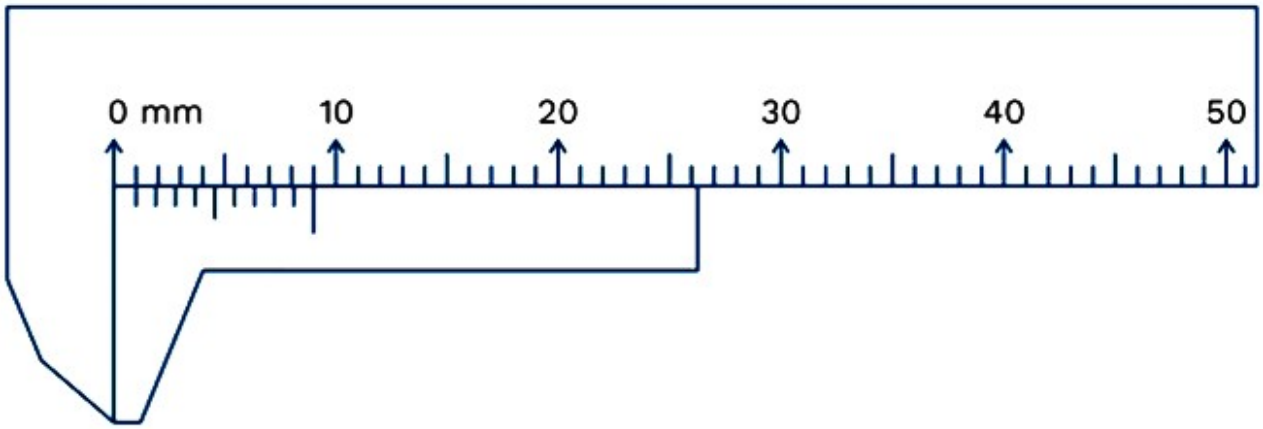
So, diameter = 14 mm + (11) (0.05 mm)

= 14.55 mm

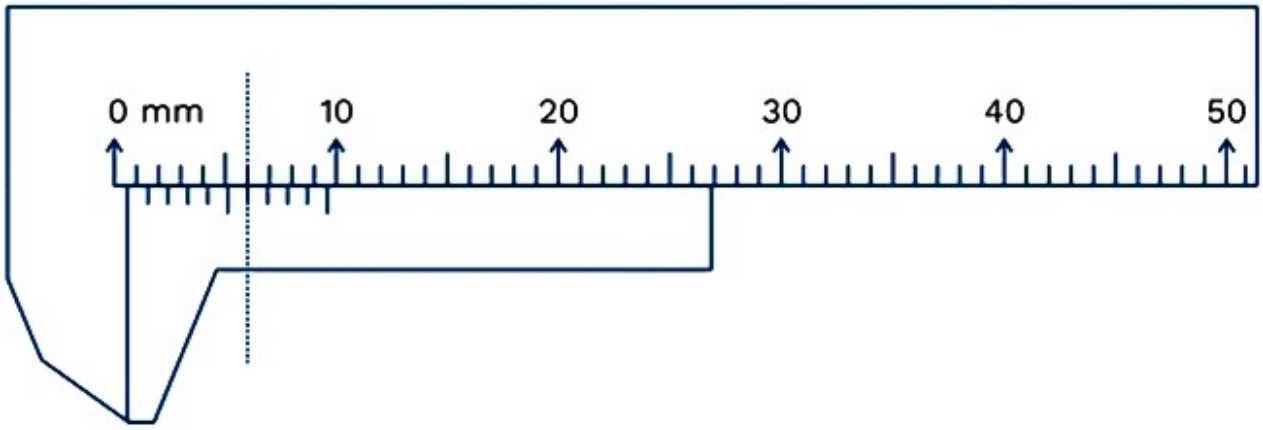
Zero Error:

If there are no objects between the tooths (i.e. the tooths are in contact), vernier will give the value zero. But due to some excess material on the two tooths, even if there is no object between the two jaws, it will give an excess reading result. This excessreading is called a zero error

Vernier calliper with no zero error



Vernier calliper with zero error

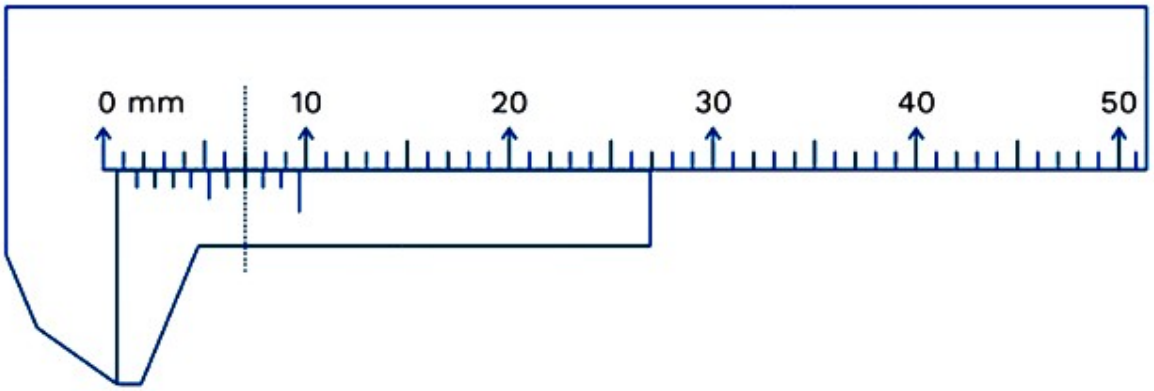


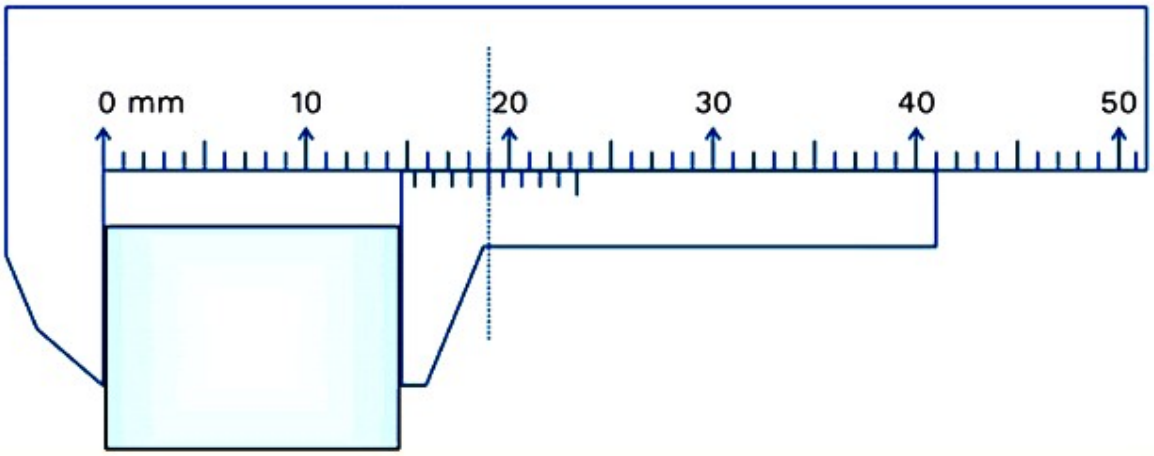
Zero error = Reading of vernier calliper when no object is placed between the jaws

= Reading of main scale + Reading of vernier scale × Least count

= 0 + 6 × 0.1 = 0.6 mm

Q. **A vernier calliper with some defect is shown in the figures in which 9 divisions on the main scale exactly match with the 10 divisions on vernier scale. What is the thickness of the object shown in the second figure measured using the defected calliper.**





Sol. As the zeroes of the main scale and the vernier scale are not matching exactly when both jaws are touching each other, so there is zero error in the instrument.

From the given figure, Zero error = 0.7 mm

From the 2nd figure, the reading of the calliper on putting the object between the jaws = 14 + 5(0.1) = 14.5 mm

In the reading of 14.5 mm, there is 0.7 mm excess reading, which has to be subtracted to get the exact value of the thickness of the object.

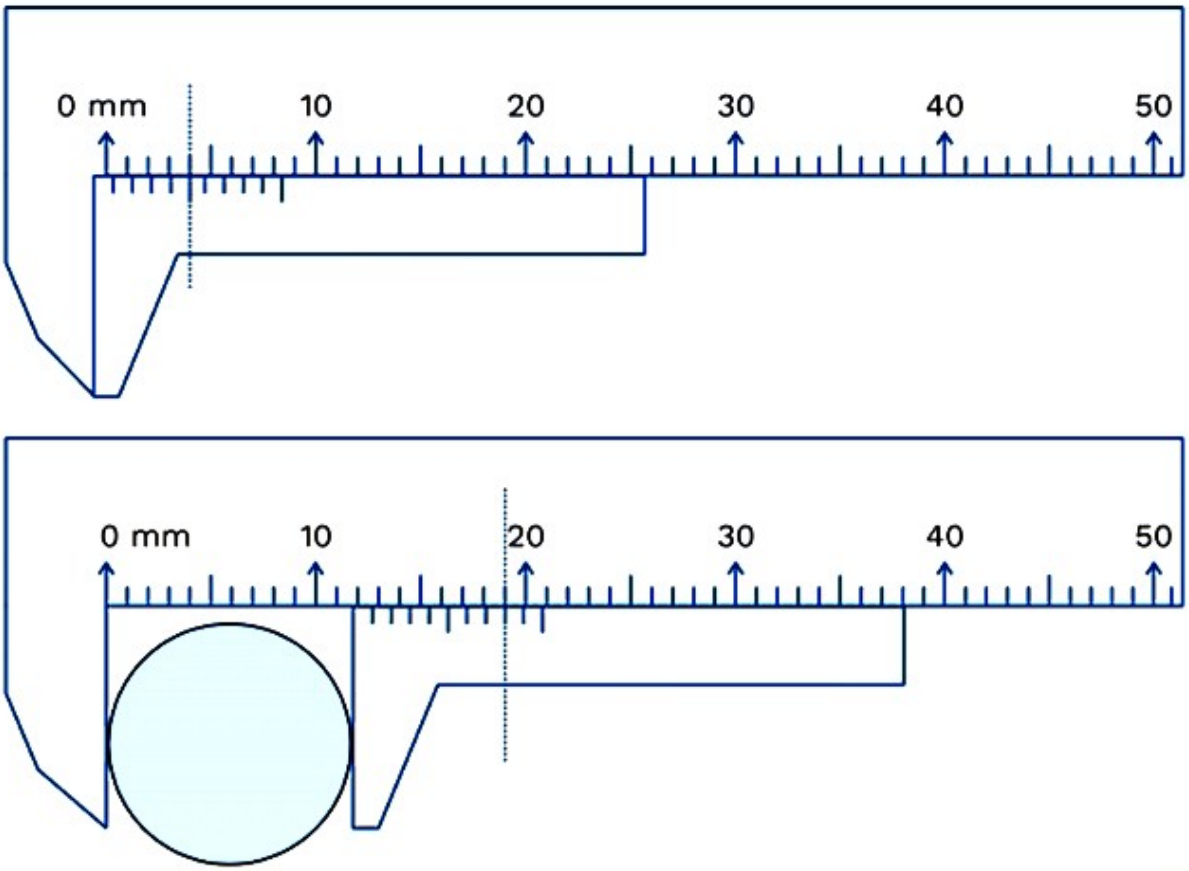
So, actual thickness = 14.5 – 0.7 = 13.8 mm.

We can also do it using the formula :

Actual reading = observed reading – excess reading (zero error)

= 14.5 – 0.7 = 13.8 mm.

Q. **Find the actual diameter of a sphere shown in the figure given below. The no object position and with object positions of the vernier calliper are given in the figure.**



Sol. Zero error = main scale reading + (vernier scale reading) (least count)

This is the case of negative zero error. To find the negative zero error in the vernier calliper, we proceed in a slightly different way.

Zero error = MSR + (VSR) (Least count)

= –1 mm + 5 (0.1 mm) = –0.5 mm (negative zero error)

From 2nd figure, we can write the Observed reading = 11.8 mm

So actual thickness = 11.8 – (–0.5) = 12.3 mm

Zero Correction

Zero correction is just the negative of the zero error.

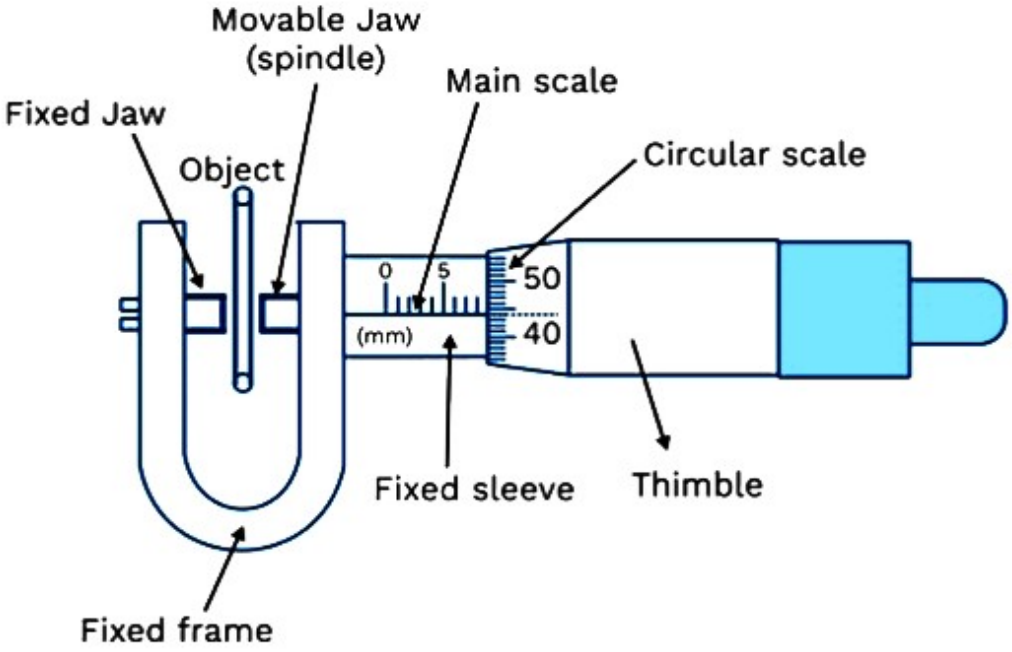
Zero correction = –(Zero error)

Actual reading = Observed reading – Excess reading (zero error)

= Observed reading + Zero correction

Screw Gauge or Micro-meter :

Description of screw gauge is shown in the figure given below.

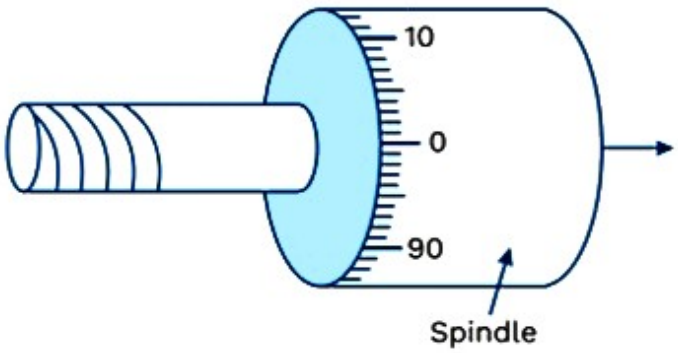


There is a fixed jaw and a movable jaw in a screw gauge. The object whose measurement is to be taken is fitted in between the jaws of the screw gauge. The fixed sleeve shown in a figure has main scale printed on it. The least count of the main scale is generally 1 mm.

A thimble is movable part on the fixed sleeve. A circular scale generally having 100 divisions is printed on the thimble. The thimble rotates on the sleeve with help of a screw mechanism.

The smallest value that the screw gauge can measure is 0.01 mm (10 µm). As the least count is of the order of microns, so screw gauge is also termed as "micrometer".

In one complete rortation of the circular scale, the screw fitted in the gauge moves forward or backward by 1 mm. It is called 'pitch'. The rotation of the circular scale is divided into 100 equal parts. So, 1 mm gets divided into 100 parts.



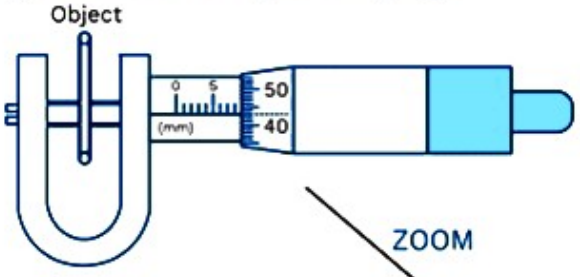
1 full rotation of circular scale = 1 mm
100 circular scale divisions = 1 mm

So, 1 circular scale division (CSD) = $\frac{1\text{ mm}}{100}$ = Least count of screw gauge

So, we can say that,

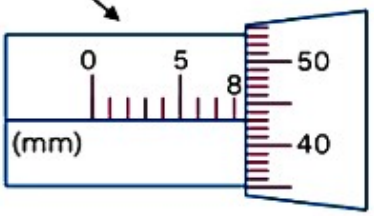
$$\text{Least count of a screw gauge} = \frac{1\text{ mm}}{100} = \frac{\text{pitch of screw}}{\text{number of divisions on circular scale}}$$

Finding thickness using screw gauge



Object is inserted between the jaws

ZOOM



1 circular part = $\frac{1\text{ mm}}{100}$

43 circular part = $43 \times \frac{1\text{ mm}}{100}$

Object thickness = 8. ------

= 8 mm + 43 circular divisions

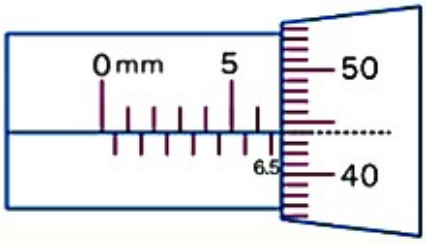
$$= 8\text{ mm} + 43\left(\frac{1\text{ mm}}{100}\right)$$

Main scale reading | Circular scale reading | Least count

$$\text{Thickness of object} = \text{Reading of screwgauge} = [\text{Main scale reading}] + [\text{Circular scale reading}] \; [\text{Least count}]$$

$$\text{Least count} = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$

Q. **In a screw gauge, main scale has $\frac{1}{2}$ mm marks. Given that in one full rotation, the screw moves forward by $\frac{1}{2}$ mm. If the circular scale has 50 divisions, then find the reading of screw gauge.**



Sol. From the given figure, the object's thickness is slightly greater than 6.5 mm.

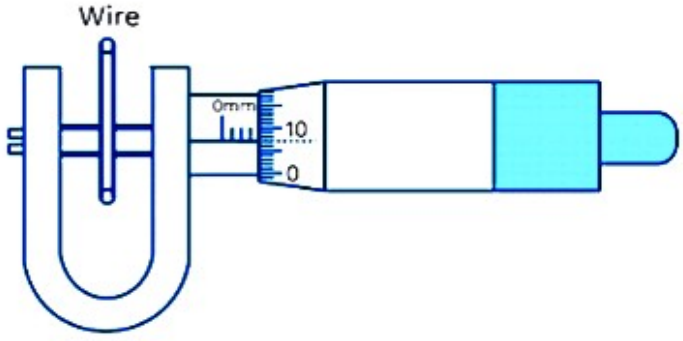
The reading of the circular scale is 44.

Least count will be, LC = (0.5 mm) / 50

So, the thickness of the object = $6.5 \text{ mm} + 44 \left(\frac{1/2 \text{ mm}}{50}\right)$

$= 6.94 \text{ mm}$

Q. **A screw gauge is being used to measure the diameter of a thin wire. The given screw gauge has a pitch of 1 mm and there are 100 divisions on the circular scale. The screw gauge with the wire fitted between its jaws is as shown in the figure.**



If the error in measurement of mass of the wire is zero and the error in the measurement of the length of the wire is 1%, then find the maximum possible error in measurement of density of the material of the wire.

Sol. $\rho = \dfrac{m}{\left(\dfrac{\pi d^2}{4}\right)\ell}$

$$\left(\frac{\Delta\rho}{\rho}\right) = 2\frac{\Delta d}{d} + \frac{\Delta\ell}{\ell}$$

Δd = least count of instrument used to measure the diameter $= \dfrac{1\text{ mm}}{100} = 0.01$ mm

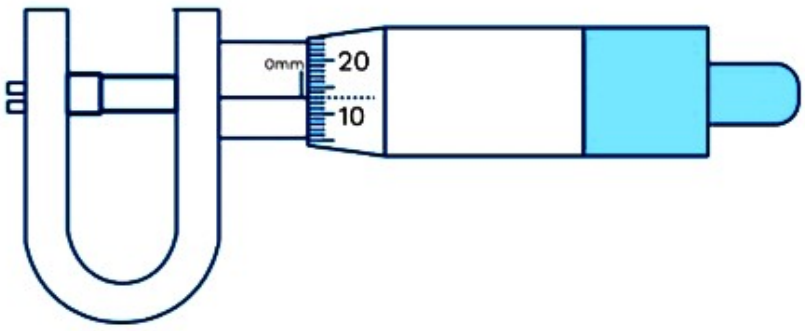
and d = 3.07 mm from the figure

$$\text{So, } \left(\frac{\Delta\rho}{\rho}\right)_{max} = \left(2 \times \frac{0.01}{3.07} + \frac{1}{100}\right) \times 100\,\%$$

$$\left(\frac{\Delta\rho}{\rho}\right)_{max} = 1.65\,\%$$

Zero Error in screw gauge :

When no object is fitted in between the jaws of a screw gauge, then the zero of the main scale and the zero of the circular scale should be coinciding.
But sometimes, the screw gauges are faulty and they show some reading even if the jaws are in contact. This reading value is called zero error.

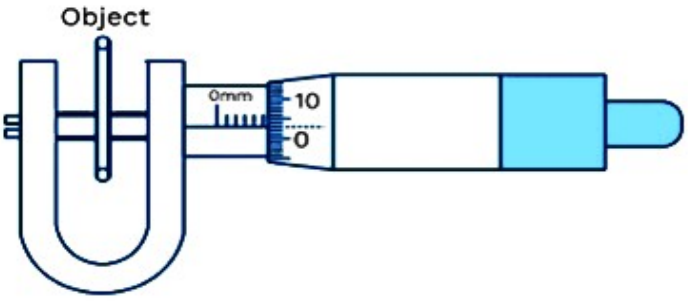


Zero error = Main scale reading + Circular scale reading × Least count

For the figure of the screw gauge shown above,

Zero error = 0 + 13 × 0.01 = 0.13 mm

Consider that we put a wire between the jaws and the figure is as shown below, then,



It gives 5.03 mm

↓

In which there is 0.13 mm excess reading which has to be subtracted

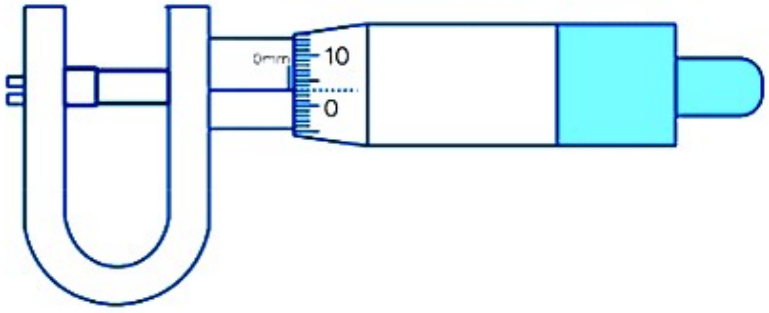
So actual thickness = 5.03 − 0.13 = 4.90 mm

Observed reading

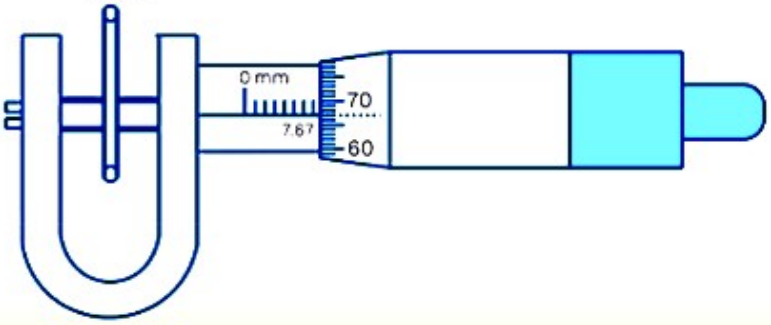
Excess reading (zero error)

Actual reading = observed reading − excess reading (zero error)

Q. **What is the thickness of the wire placed between the jaws of a screw gauge if the two conditions of the screw gauge are shown in the figure below. Consider 100 divisions on the circular scale and a pitch of 1 mm for the screw gauge.**

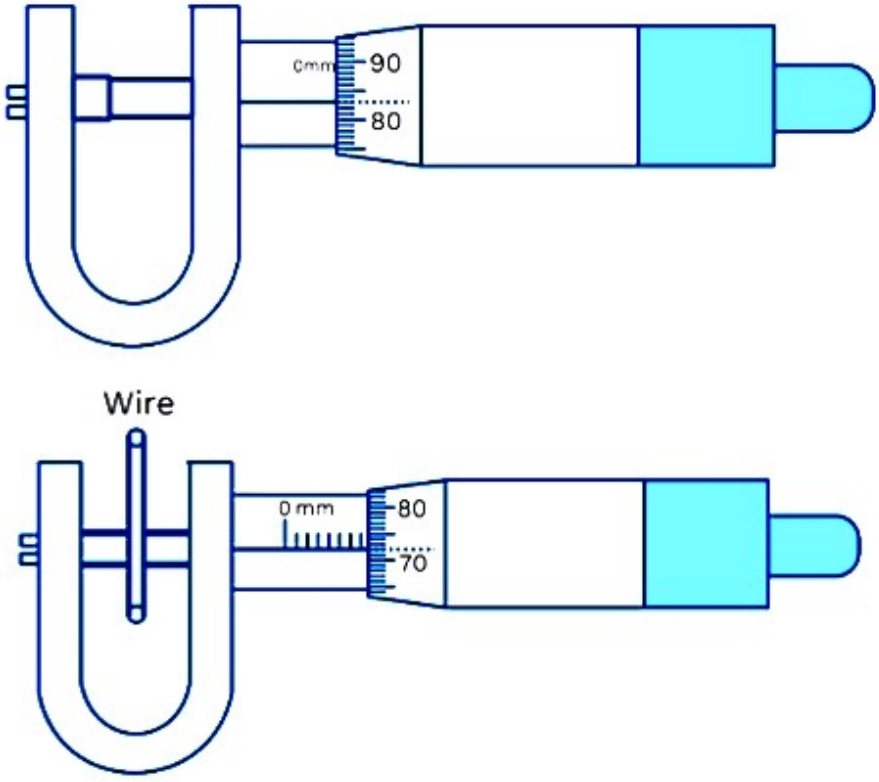


Wire



Sol. In the first figure, 3rd marking of the circular scale is coinciding with zero of the main scale. So,
Zero error = 0.03 mm.
In the 2nd figure, it is showing a reading of 7.67 mm out of which 0.03 mm is excess reading. So, it has to be subtracted to get the final reading.
Then, actual reading = 7.67 – 0.03 = 7.64 mm.

Q. **What is the thickness of the wire placed between the jaws of a screw gauge if 83rd circular division is matching with zero of main scale when no wire is placed between the jaws. Consider 100 divisions on the circular scale and a pitch of 1 mm for the screw gauge.**



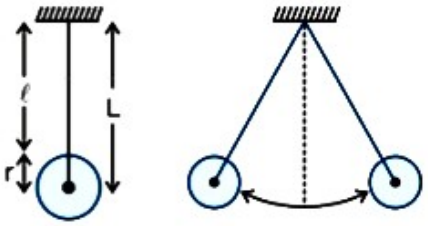
Sol. This is a case of negative zero error.
Zero error = Main scale reading + Circular scale reading × Least count
= (–1 mm) + (83) × (0.01) = –0.17 mm
From the 2nd figure, the reading of the screw gauge is 7.72 mm.
In the reading 7.72 mm, –0.17 mm is excess reading. So, we have to subtract it to get the actual thickness of the wire.
Actual wire thickness = 7.72 – (–0.17) = 7.89 mm

2 Measurement of g using a Simple Pendulum

Measurement of g using a Simple Pendulum

A simple pendulum is an arrangement of a small bob attached to a cotton thread whose one end is fixed to a point.

On given a small angular displacement ot the bob, it starts oscillating as shown in the figure below.



To find time period of oscillation of the pendulum experimentally, we measure the time taken to complete 50 or 100 oscillations using a stopwatch.

Then, time period for one complete oscillation can be found as,

$$T = \text{time period of the simple pendulum} = \frac{\text{Time taken for 50 oscillations}}{50}$$

The theoretical formula for the time period of simple pendulum is, $T = 2\pi\sqrt{\frac{L}{g}}$

We can rearrange and write this formula as,

$$\Rightarrow g = 4\pi^2 \frac{L}{T^2} \qquad \ldots(1)$$

Where L = equivalent pendulum length = length of thread (l) + radius (r) of bob,

As T and L become known, we can easily find 'g' using equation (1).

Graphical method for determination of 'g'

Equation (1) can be rearranged to write,

$$T^2 = \left(\frac{4\pi^2}{g}\right)L \qquad \ldots(2)$$

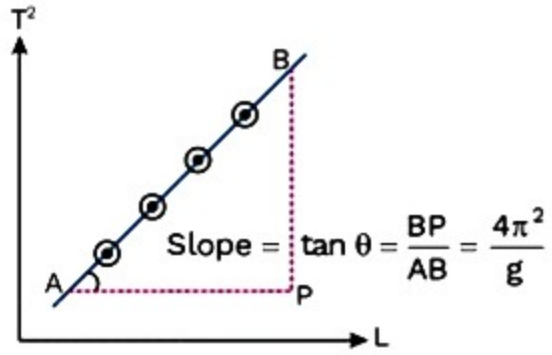
So, $T^2 \propto L$

We can calculate the time period T for different values of pendulum length L.

Then, we have to plot the T^2 v/s L graph.

As per equation (2), the graph should be a straight line, having slope $= \left(\frac{4\pi^2}{g}\right)$.

Now, we have to find the slope of T^2 v/s L graph and equate it to $\left(\frac{4\pi^2}{g}\right)$ We can then simplify the expression to find the value of "g".



Q. **The length of the string in an experiment is 23.2 cm while the radius of the bob used is 1.32 cm. Find the value of g in suitable significant figures if the time taken for 10 oscillations by the pendulum is 10.0 s. (take $\pi^2 = 10$)**

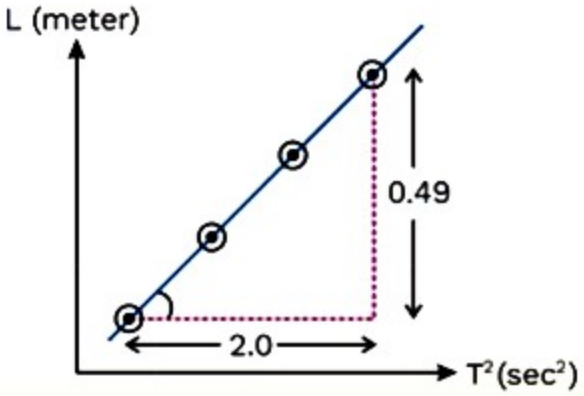
Sol. Equivalent pendulum length, L = 23.2 cm + 1.32 cm = 24.5 cm (upto 3 significant figures)

And time-period $T = \frac{10.0}{10} = 1.00\,s$

$$g = 4\pi^2 \frac{L}{T^2} = 4 \times 10 \frac{24.5\,cm}{(1.00)^2}$$

$$= 4 \times 10 \times \frac{24.5 \times 10^{-2} m}{(1.00)^2\,sec^2} = 9.80\,m/sec^2$$

Q. **A graph between equivalent pendulum length L and square of the time period T^2 is plotted for a simple pendulum. Find the value of g from the graph. (take $\pi^2 = 10$)**



Sol. Using the expression, $L = \left(\frac{g}{4\pi^2}\right)T^2$

Slope of the given graph will be $\left(\frac{g}{4\pi^2}\right)$

$$\text{Slope} = \frac{0.49}{2} = \frac{g}{4\pi^2} \quad \Rightarrow g = 9.8 \text{ m/sec}^2$$

The major errors in this experiment are

(a) (a) Systematic: Error due to finite pendulum amplitude (because of incorrect SHM motion). This can be corrected by using exact numerical estimates for the Time period. However, the effort is to keep the amplitude low.

(b) Statistical: Errors which arise from the measurement of length and time.

$$\frac{\delta g}{g} = \frac{\delta L}{L} + 2\left(\frac{\delta T}{T}\right)$$

The contributions to δL, δT are both systematic and statistical. They are reduced by the averaging process. The systematic error in L are reduced by plotting of several values of L relative to T2and fitting them to a straight line. The slope of this curve gives the correct value of L/T^2

Maximum permissible error in 'g' due to error in measurement of ℓ, r and T

$$g = 4\pi^2 \frac{L}{T^2}, \quad (L = \ell + r,\ T = t/50)$$

$$g = 4\pi^2 \frac{(\ell + r)}{(t/50)^2} = 4\pi^2(2500)\frac{\ell + r}{t^2}$$

$$\ln g = \ln 4\pi^2(2500) + \ln(\ell + r) - 2\ln(t)$$

$$\left(\frac{\Delta g}{g}\right)_{max} = \frac{\Delta\ell + \Delta r}{\ell + r} + 2\frac{\Delta t}{t}$$

Q. **The length of the string in an experiment is 23.2 cm while the radius of the bob used is 1.32 cm. Find the maximum permissible error in measurement of g if the time taken for 10 oscillations by the pendulum is 10.0 s. (take $\pi^2 = 10$)**

Sol.

$\ell = 23.2$ cm $\rightarrow$ $\Delta\ell = 0.1$ cm

$r = 1.32$ cm $\rightarrow$ $\Delta r = 0.01$ cm

$t = 10.0$ sec $\rightarrow$ $\Delta t = 0.1$ sec

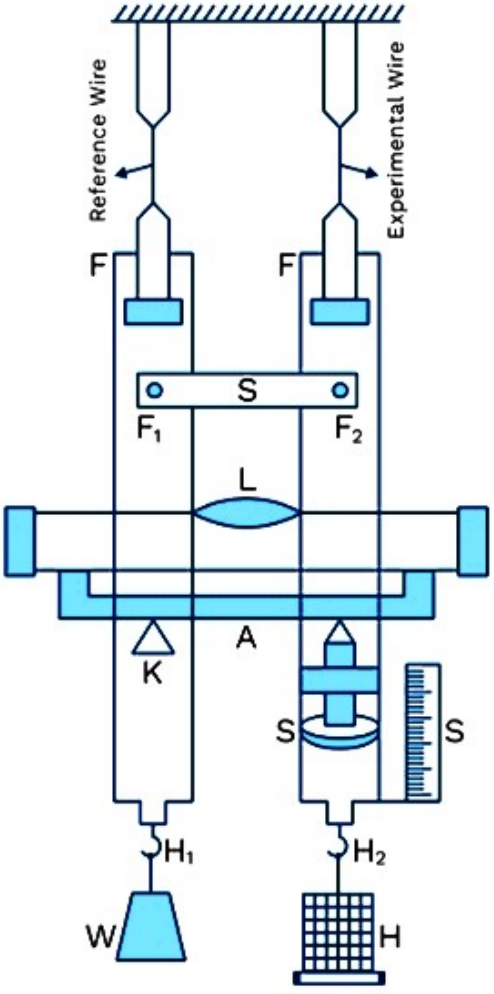
$$\left(\frac{\Delta g}{g}\right)_{max} = \left(\frac{0.1 \text{ cm} + 0.01 \text{ cm}}{23.2 \text{ cm} + 1.32 \text{ cm}} + 2\frac{0.1 \text{ sec}}{10.0 \text{ sec}}\right) \times 100\% = 2.44\%$$

3 Young's Modulus by Searle's Method

Searle's Apparatus (Static Method):

In the experimental set up, two wires (one reference wire and other the test wire) are fitted on a frame allowing only vertical relative motion in the wires.

A spirit level is fitted on a cross bar frame. The other end of the spirit level rests on a tip of a micrometer screw. If the test wire elongates on applying load on it, the spirit level tilts and this elongation can be measured by the scale on the micormeter.



Procedure :

(i) First of all measure the length of the test (experimental) wire.

(ii) Measure the diameter of the experimental wire usinig a screw gauge.

(iii) Adjust the micrometer such that the spirit level is horizontal and note the reading of the micrometer scale.

(iv) Now, increase the load on the experimental wire by adding weights in stepd of 0.5 kg in the load box H. Note the reading on the micrometer scale as the experimental wire elongates.

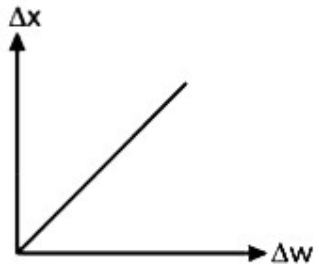
(v) Now, unload the weights from the load box H in the same fashion and take the readings of the micrometer scale while unloading.

Both the values taken during loading and unloading should match closely.

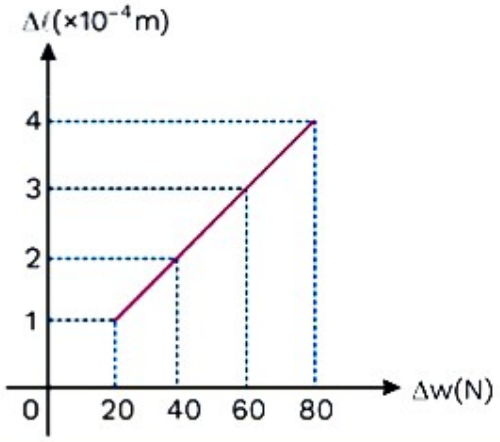
If an extra weight Δw is added to the wire, then extra elongation Δx in the wire is directly proportional to the extra weight added.

$$x = \frac{\ell_0}{\pi r^2 Y} w \Rightarrow \Delta x = \frac{\ell_0}{\pi r^2 Y}(\Delta w)$$

Thus the graph between Δx and Δw has slope $= \left(\frac{\ell_0}{\pi r^2 Y}\right)$



Q. **Find the Young's modulus of a wire whose Δx vs Δw graph is shown in the figure below. Assume the initial length of the wire to be 1 m and a cross sectional area of 10^{-6} m².**



Sol. $\Delta\ell = \left(\frac{\ell_0}{AY}\right)\Delta w$, slope $= \frac{\ell_0}{AY} = \frac{1\times 10^{-4}}{20} \Rightarrow \frac{1}{(10^{-6})Y} = \frac{1\times 10^{-4}}{20}$

$Y = 20 \times 10^{10} = 2 \times 10^{11}\ N/m^2$

Maximum permissible error in 'Y'

We can write the formula for Young's modulus to be, $Y = \frac{\ell_0}{\pi r^2 x} mg$

For no error in the measurement of mass; maximum permissible error in Y is given by

$$\left(\frac{\Delta Y}{Y}\right)_{max} = \frac{\Delta \ell_0}{\ell_0} + 2\frac{\Delta r}{r} + \frac{\Delta x}{x}$$

Q. **In Searle's experiment to find Young's modulus, following data is given:**
Wire diameter, d = 0.050 cm,
Length of wire, l = 125 cm
On applying a load of m = 20.0 kg, extension in the length of wire is 0.100 cm.
What is the maximum permissible error in Young's modulus (Y).
Use : $Y = \frac{mg\ell}{(\pi/4)d^2x}$**. Least count for mass, length, diameter and extension measurement are respectively 0.1 kg, 1 cm, 0.001 cm and 0.001 cm.**

Sol. $\frac{mg}{\pi d^2/4} = Y\left(\frac{x}{\ell}\right) \Rightarrow Y = \frac{mg\ell}{(\pi/4)\,d^2x}$

$$\left(\frac{\Delta Y}{Y}\right)_{max} = \frac{\Delta m}{m} + \frac{\Delta \ell}{\ell} + 2\frac{\Delta d}{d} + \frac{\Delta x}{x}$$

$\Delta m = 0.1 kg;\ \Delta \ell = 1 cm;\ \Delta d = 0.001 cm;\ \Delta x = 0.001 cm$

$$\left(\frac{\Delta Y}{Y}\right)_{max} = \left(\frac{0.1 kg}{20.0 kg} + \frac{1 cm}{125 cm} + \frac{2 \times 0.001 cm}{0.05 cm} + \frac{0.001 cm}{0.100 cm}\right) \times 100\% = 6.3\%$$

4 Specific Heat Capacity of a Liquid using Calorimeter

A known mass (m) of liquid is heated in a calorimeter of known mass (M) by supplying a current I during a time t. If V is the potential drop across the heating coil, T_0 is the initial temperature of the liquid and temperature of the liquid reaches to a maximum value of T_2

Then by calorimetry, we can write,

$m\ S_l\ (T_2 - T_0) + M\ S_c\ (T_2 - T_0) = I.V.t$

or $m\ S_l + M\ S_c = I.V.t\ /\ (T_2 - T_0)$ ...(1)

where, S_c is the specific heat of the calorimeter material. If S_c is known, then we can calculate S_l.

Q. **In this experiment voltage across the heater is 100.0 V and current is 10.0A, and heater was switched on for t = 700.0 sec. Initially all elements were at room temperature θ_0 = 10.0°C and final temperature was measured as θ_f = 73.0°C. Mass of empty calorimeter was 1.0 kg and the combined mass of calorimeter + liquid is 3.0 kg. The specific heat capacity of the calorimeter S_c = 3.0 × 10^3 J/kg°C. The fall in temperature 350 second after switching off the heater was 7.0°C. Find the specific heat capacity of the unknown liquid in proper significant figures.**

(A) 3.5 × 10^3 J/kg°C **(B) 3.50 × 10^3 J/kg°C**

(C) 4.0 × 10^3 J/kg°C **(D) 3.500 × 10^3 J/kg°C**

Sol. Corrected final temperature = θ_f = 73.0°C + 7.0°C = 80°C

$$S_\ell = \frac{\frac{(100.0)\ (10.0)\ (700.0)}{80.0 - 10.0} - (1.0)\ (3.0 \times 10^3)}{3.0 - 1.0}$$

$= 3.5 \times 10^3$ J/kg°C

Q. **If mass and specific heat capacity of calorimeter is negligible, what would be maximum permissible error in S_ℓ. Use the data mentioned below.**

$m_1 \to 0$, $S_c \to 0$, m_2 = 1.00 kg, V = 10.0 V, I = 10.0 A, t = 1.00 × 10^2 sec., θ_0 = 15°C, Corrected θ_f = 65°C

(A) 4% **(B) 5%**

(C) 8% **(D) 12%**

Sol. If $m_1 \to 0$, $S_c \to 0$

$$S_\ell = \frac{VIt}{m_2(\theta_f - \theta_0)}$$

$$\frac{\Delta S_\ell}{S_\ell} = \frac{\Delta V}{V} + \frac{\Delta I}{I} + \frac{\Delta t}{t} + \frac{\Delta m_2}{m_2} + \frac{\Delta\theta_f + \Delta\theta_0}{\theta_f - \theta_0} = \frac{0.1}{10.0} + \frac{0.1}{10.0} + \frac{0.01 \times 10^2}{1.00 \times 10^2} + \frac{0.01}{1.00} + \frac{1+1}{50} = 8\%$$

5 u-v Method for Focal Length

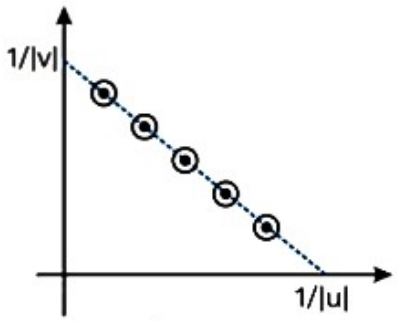
Focal length calculation for a concave mirror and a convex lens using the u-v method

An optical bench with a concave mirror or convex lens is used to perform this experiment. A scale is fitted on the optical bench to measure various lengths along the principal axis of the mirror or the lens.

An object such as a candle is placed at a certain distance "u" from the lens and distance of its real image "v" is measured by placing a screen at the position of image formation.

For various values of u, we get different values of v. On plotting a graph between u and v, we get a rectangular hyperbolic curve.

On plotting a graph between $\frac{1}{|v|}$ and $\frac{1}{|u|}$, we get a straight line.



The intercepts of the graph are equal to $\frac{1}{|f|}$, where f is the focal length.

Error : Due to improper positioning of the holder, there may be systematic errors in the experiment. On reversing the holder position, we can eliminate the systematic errors.

From the lens formula, we can write the error equation as,

$$\frac{\delta f}{f^2} = \frac{\delta u}{u^2} + \frac{\delta v}{v^2}$$

Q. **In u – v method to find focal length of a concave mirror, if object distance was found to be 10.0 cm and image distance was also found to be 10.0 cm then find maximum permissible error in f, due to error in u and v measurement. (Least count of the scale on optical bench is 1 mm.)**

Sol. $\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \Rightarrow \frac{1}{(-10)}+\frac{1}{(-10)}=\frac{1}{f} \Rightarrow |f| = 5\,cm$

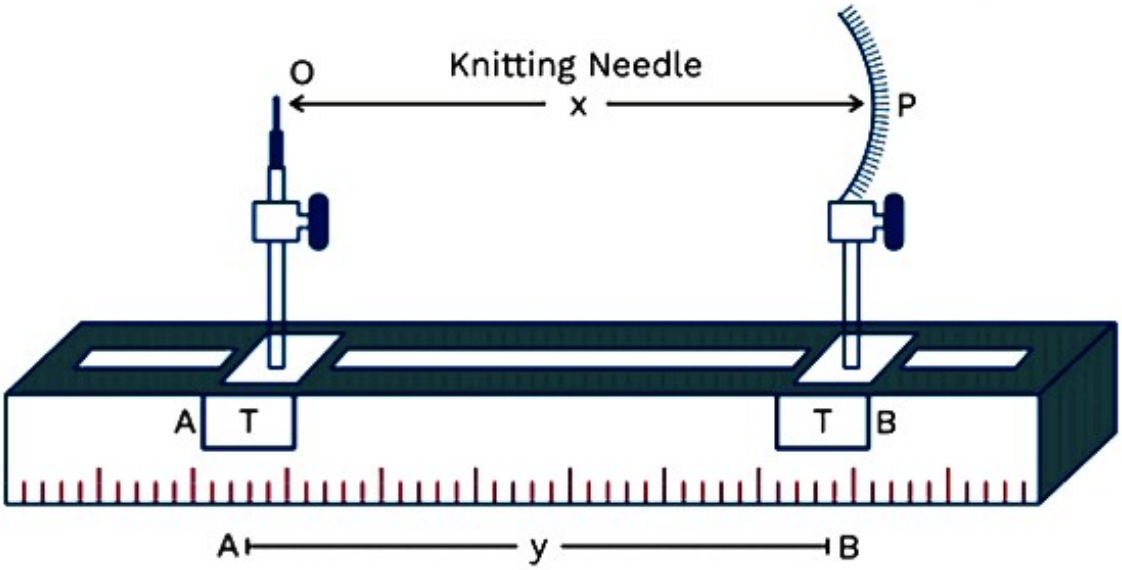
On differentiating the mirror formula,

$$(\Delta f)_{max} = \left(\frac{\Delta u}{u^2}+\frac{\Delta v}{v^2}\right)\times f^2 \Rightarrow (\Delta f)_{max} = \left(\frac{0.1}{10^2}+\frac{0.1}{10^2}\right)\times 5^2 = 0.05\ cm$$

So, $f = (5 \pm 0.05)\,cm$

INDEX ERROR

In the application of the mirror formula/lens formula, the distances of the object and the image from the pole of the mirror or from the optical centre of the lens are required. But actually we put the distances between the stands or indices which slide on the table. These distances are not the exactly required distances and this slight mismatch causes index error in the observations which is constant for every observation taken during the experiment.



Determination of index correction

We can find the index error as,

Index error = Observed distance – Actual distance

To find the index error, a knitting needle can be used by fitting it in between the pole and the object. The length of the needle used gives the actual distance.

Then, index error, e = Observed distance – Actual distance

= Separation between indices A and B – Length of knitting needle = y – x

After finding e for every observation, we can write,

Actual distance = Observed distance (Separation between the indices) – Excess reading (e)

One more term, Index correction is just the opposite of index error.

Index correction = – Index error

Q. **Find the index error and index correction for object distance u.**
Length of knitting needle between pole and object = 20.0 cm
Separation between the indices = 20.2 cm.

Sol. Index error = Observed reading – Actual reading = 20.2 – 20.0 = 0.2 cm

Index correction = -0.2 cm

Q. **Find the index error and index correction for image distance v.**
Length of knitting needle between pole and object = 20.0 cm
Separation between the indices = 19.9 cm.

Sol. Index error, e = 19.9 cm − 20.0 cm = −0.1 cm.

Index correction = 0.1 cm

Q. **For index correction of previous two questions, find focal length of the concave mirror when object distance u is observed to be 30.2 cm and the image distance v is observed to be 19.9 cm.**

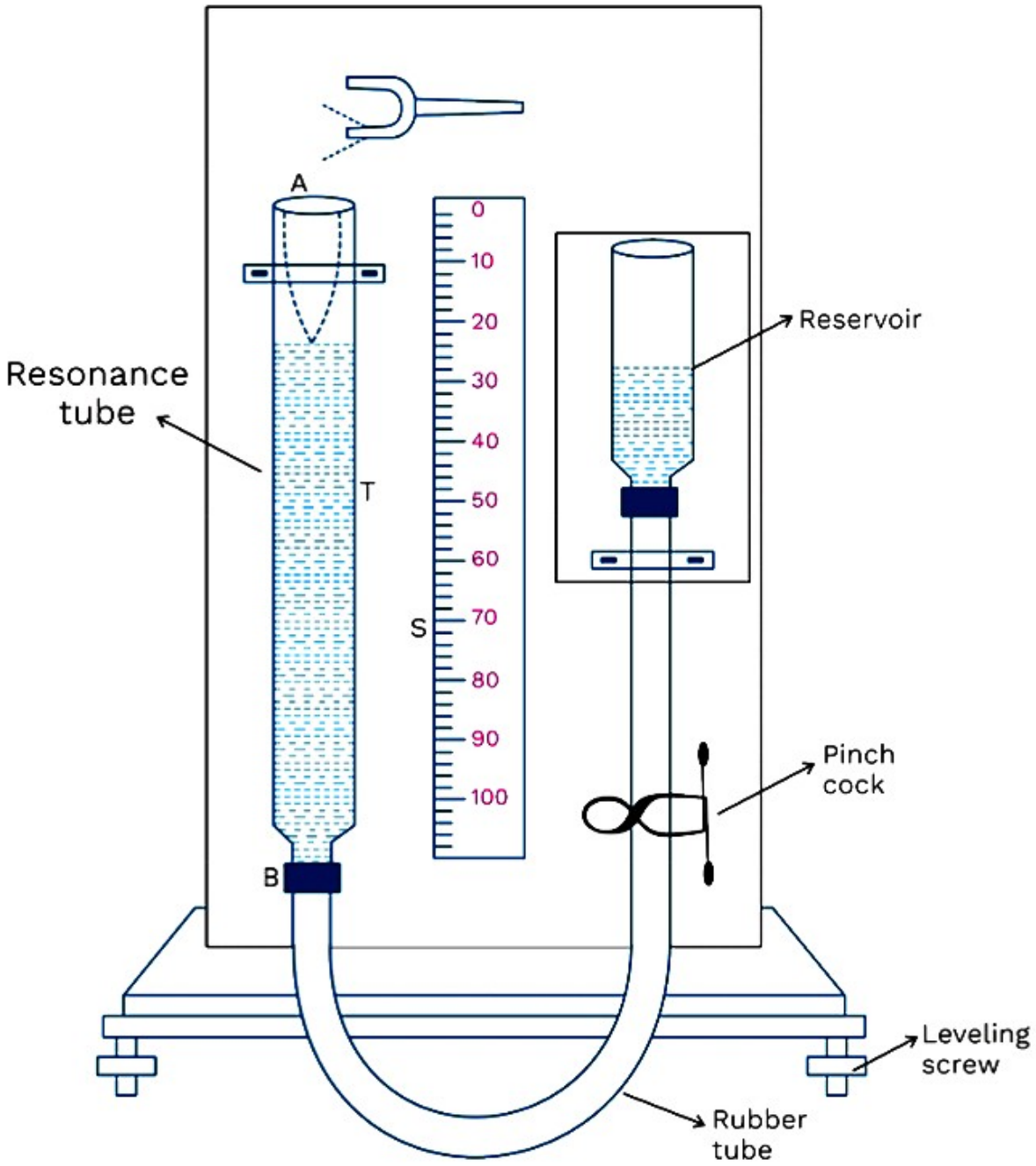
Solz. u = 30.2 − 0.2 = 30.0 cm

v = 19.9 − (−0.1) = 20.0 cm

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow f = 12.0\,cm$$

6 Resonance Tube Experiment

Determining speed of sound using resonance tube method



An experimental set up of resonance tube to find the speed of sound in air is shown in the figure shown above. The level of water in the resonance tube can be varied by supplying or taking water from or in the reservoir.

Principle: Resonance tube is similar to a closed organ pipe.

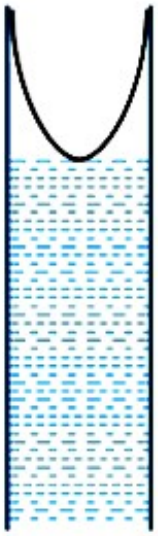
So, the frequencies of resonance are given as,

$$f_n = (2n-1)\frac{V}{4\ell_{eq}}, \text{ where } n = 1, 2, 3, 4, ...$$

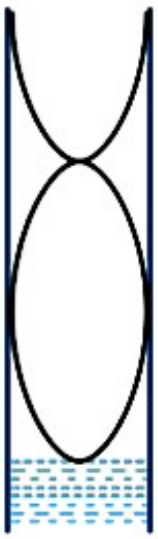
By using a tuning fork of frequency fo , natural frequency of the resonance tube and the frequency of the fork must be same for their resonance.

$$(2n-1)\frac{V}{4\ell_{eq}} = f_0 \Rightarrow \ell_{eq} = (2n-1)\frac{V}{4f_0}$$

For the first resonance $\ell_{eq} = \frac{V}{4f_0} =$ (corresponding to 1st mode)



For the second resonance $\ell_{eq} = \frac{3V}{4f_0} =$ (corresponding to 2nd mode)



Working : The length of the resonance tube is 100 cm and it is filled with water initially. Water level can be reduced by taking some water to the reservoir in order to increase the length of the air column.

On using a tuning fork of frequency fo at the top of the air column, if first resonance is observed at a length l1 , then,

$$\ell_{eq_1} = \frac{V}{4f_0} \Rightarrow \ell_1 + e = \frac{V}{4f_0} \qquad ...(1)$$

Where e is end correction

On further reducing the level of water to increase the length of the air column, if next resonance is observed at a length l_2, then,

$$\ell_{eq_2} = \frac{3V}{4f_0} \Rightarrow \ell_2 + e = \frac{3V}{4f_0} \qquad ...(2)$$

Subtracting (1) from (2), we get,

$$V = 2f_0(\ell_2 - \ell_1)$$

Using $\ell_3 + e = \frac{5V}{4f_0}$, we can write,

$$\ell_3 = 2\ell_2 - \ell_1$$

Also, end correction, $e = \frac{\ell_2 - 3\ell_1}{2}$

Generally, e = 0.3 d (where d = diameter of tube)

Observation Table

		Position of water level (cm)			
Frequency of tuning for in (Hz) (f_0)	Resonance	Water level is falling	Water level is rising	Mean resonant length	Speed of sound $V = 2f_0(\ell_2 - \ell_1)$
340 Hz	1st Resonance	23.9	24.1	$\ell_1 = 24.0$	V = ...
	2nd Resonance	73.9	74.1	$\ell_2 = 74.0$	

Calculations:

ℓ_1 = 24.0 cm ; ℓ_2 = 74.0 cm

$v = 2f_0(\ell_2 - \ell_1) = 2(340)(0.740 - 0.240) = (2)(340)(0.500) = 340$ m/sec.

$$\ell_1 + e = \frac{V}{4f_0} \quad \text{and} \quad \ell_2 + e = \frac{3V}{4f_0}$$

Solve both equations and get e = 1 cm

For third resonance

$$\ell_3 + e = \frac{5V}{4f_0} \Rightarrow \text{get } \ell_3 = 124\,\text{cm}$$

Also, diameter of tube, $d = \frac{e}{0.3} = \frac{1}{0.3}\text{cm} = 3.3\,\text{cm}$

Maximum Permissible Error in speed of sound due to error in f_0, ℓ_1, ℓ_2

We found the speed of sound as,

$V = 2f_0(\ell_2 - \ell_1)$

$\ln V = \ln 2 + \ln f_0 + \ln(\ell_2 - \ell_1)$

Maximum permissible error in speed of sound $= \left(\frac{\Delta V}{V}\right)_{max} = \frac{\Delta f_0}{f_0} + \frac{\Delta\ell_2 + \Delta\ell_1}{(\ell_2 - \ell_1)}$

Q. **1st and 2nd resonance lengths of 224.0 cm and 74.0 cm are observed when a tuning fork of frequency (340 Hz ± 1%) is used in a resonance tube experiment. If the least count of the scale used to measure the resonance lengths is 1 mm, then find the maximum permissible error in the speed of sound.**

Sol. $\Delta\ell_1 = 0.1\text{cm}$, $\Delta\ell_2 = 0.1\text{cm}$

$$f_0 = (340\,\text{Hz} \pm 1\%) \qquad \frac{\Delta f_0}{f_0} = 1\% = \frac{1}{100}$$

$$\left(\frac{\Delta V}{V}\right)_{max} = \frac{\Delta f_0}{f_0} + \frac{\Delta\ell_2 + \Delta\ell_1}{\ell_2 - \ell_1} = \frac{1}{100} + \frac{0.1 + 0.1}{74.0 - 24.0} = \frac{1}{100} + \frac{0.2}{50.0} = 0.014$$

Now, $V = 2f_0(\ell_2 - \ell_1)$

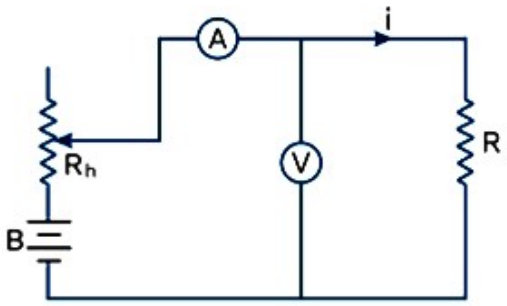
$= 2 \times 340\,(0.740 - 0.240) = 340$ m/s.

$(\Delta V)_{max} = 0.014\,V = 0.014 \times 340 = 4.76$ m/s.

7 Verification of Ohm's Law

Verification of Ohm's law using voltmeter and ammeter

A circuit is used to verify Ohm's law which contains a resistance R connected to a battery with a variable resistance (rheostat) Rh. The voltmeter and the ammeter are connected as shown in the circuit diagram below.



During the experiment, the readings of the ammeter and the voltmeter are taken by changing the value of the rheostat.

A graph is plotted between the value of the potential drop V (the voltmeter reading) and the current i (ammeter reading) and this graph is a linear plot (with some errors).

Then, we can find the resistance R by finding the slope of the V-i graph.

$$V = i\,R.$$

Errors :

The error equation for this experiment can be written as,

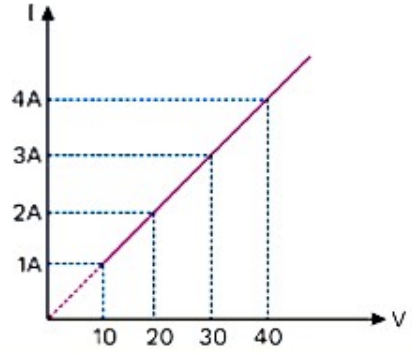
$$\left|\frac{\delta R}{R}\right| = \left|\frac{\delta V}{V}\right| + \left|\frac{\delta i}{i}\right|$$

Where δV is the least count of the voltmeter and δi is the least count of the ammeter.

Error analysis :

Where δV and δI are of the order of the least counts of the instruments used.

Q. **For the given I-V graph, if emf of battery is 100V, then what was the resistance of Rheostat adjusted at 2nd reading (I = 2A, V = 20V).**



Sol. From the curve slope $= \frac{I}{V} = \frac{1}{R} = \frac{1}{10} \Rightarrow R = 10\,\Omega$

For second reading, $I = \frac{Emf}{R + R_{rh}} \Rightarrow 2 = \frac{100}{10 + R_{rh}} \Rightarrow R_{rh} = 40\,\Omega$

Maximum possible error in resistivity

Resistivity of the wire can be calculated as:

$$\rho = \frac{RA}{L} = \frac{\pi D^2}{4L}\frac{V}{I}$$

$$\ln \rho = \ln\frac{\pi}{4} + 2\ln D - \ln L + \ln V - \ln I$$

$$\frac{d\rho}{\rho} = 2\frac{dD}{D} - \frac{dL}{L} + \frac{dV}{V} - \frac{dI}{I}$$

$$\frac{\Delta\rho}{\rho} = \pm 2\frac{\Delta D}{D} \mp \frac{\Delta L}{L} \pm \frac{\Delta V}{V} \mp \frac{\Delta I}{I}$$

$$\left(\frac{\Delta\rho}{\rho}\right)_{max} = \text{max of}\left(\pm 2\frac{\Delta D}{D} \mp \frac{\Delta L}{L} \pm \frac{\Delta V}{V} \mp \frac{\Delta I}{I}\right)$$

$$\left(\frac{\Delta\rho}{\rho}\right)_{max} = +2\frac{\Delta D}{D} + \frac{\Delta L}{L} + \frac{\Delta V}{V} + \frac{\Delta I}{I} = \text{maximum permissible error in } \rho.$$

Q. **Find the maximum possible error in measurement of resistivity of a wire when its length is 10.0 cm and its diameter is 2.50 mm. Current in the wire is measured to be 1.00 A on applying a potential difference of 10.0 V.**

Sol. $\left(\frac{\Delta\rho}{\rho}\right)_{max} = 2\left(\frac{0.01}{2.50}\right) + \left(\frac{0.1}{10.0}\right) + \left(\frac{0.1}{10.0}\right) + \left(\frac{0.01}{1.00}\right) = 3.8\,\%$

8 Meter Bridge

Meter bridge is used to find the value of an unknown resistance by balancing the bridge circuit just like a wheatstone bridge.
A resistance wire of length 1 m is attached between points A and C if the cicuit.
A jockey is moved along the length of the wire.
R is the unknown resistance and S is a known resistance.
At the balanced condition of the bridge,

$R(100 - l) = S(l)$

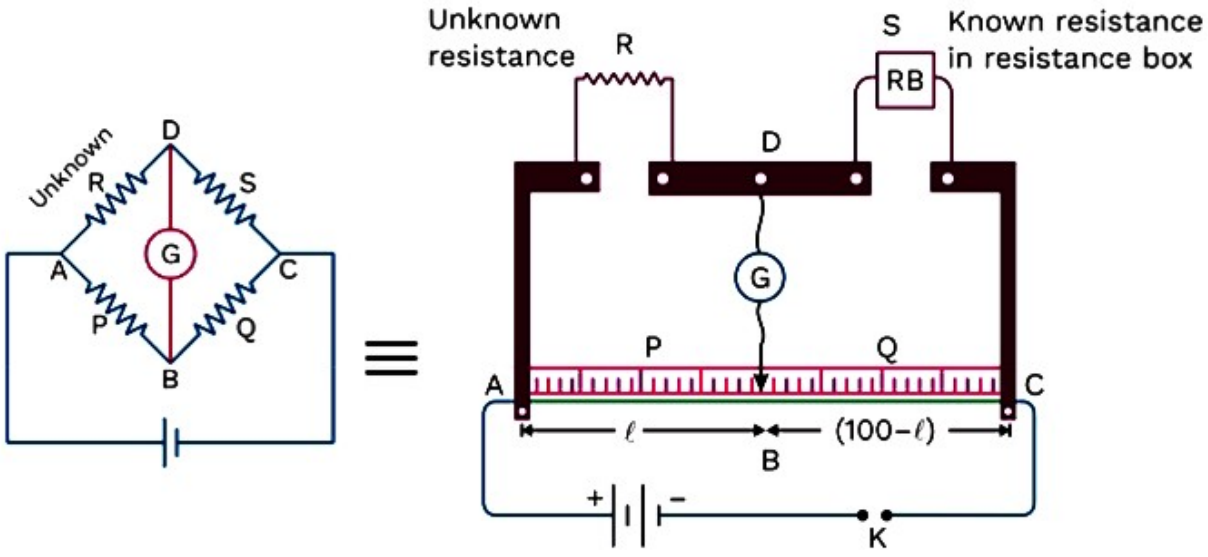
At this condition, the reading of the galavanometer is zero and it shows no deflection.

Then, we can find the value of the unknown resistance as, $R = S\dfrac{l}{100 - l}$ ...(1)

For a length L and diameter d of the wire, resistivity is

$$\rho = \frac{R\left(\frac{\pi d^2}{4}\right)}{L}$$

By equation (1), $\rho = \dfrac{\pi d^2}{4L}\left(\dfrac{l}{100 - l}\right)S$



Q. **Find the resistivity of a wire having a length of 31.4 cm and a diameter of 1 mm used as unknown resistance in a meter bridge experiment. Given that the known resistance in the experiment is 300 ohms and the null point is obtained ad a distance of 25.0 cm from end A.**

Sol. $\dfrac{R}{300} = \dfrac{25}{75} \Rightarrow R = 100 \Rightarrow \rho = \dfrac{R\pi d^2}{4L} = 2.5 \times 10^{-4}\ \Omega\text{-m}$

Q. **No deflection point is observed to be 33.7 cm in a meter bridge circuit. On shunting the resistance S by a 12 Ω resistance, the null point shifts by 18.2 cm. Find the value of the unknown resistance.**

Sol. $\frac{R}{S} = \frac{33.7}{100 - 33.7}$ and $\frac{R}{\left(\frac{12 \times S}{12 + S}\right)} = \frac{(33.7 + 18.2)}{100 - (33.7 + 18.2)}$

On solving we get, R = 6.86 Ω

End Corrections

The end points A and C are fitted on a board with some metallic strips. So, in actual practice, there is some additional resistance due to these metallic strips also. For accurate result of the experiment, the correction for these additional resistances should be considered.
So, we add some additional lengths (α and β) at the ends for exact results.
Then l is replaced by l + α and 100 − l is replaced by 100 − l + β.
We can find the values of α and β by using two known resistances at R = R_1 and S = R_2. If a null point is obtained as a length l_1, then,

$$\frac{R_1}{R_2} = \frac{\ell_1 + \alpha}{100 - \ell_1 + \beta} \quad \ldots(1)$$

Now, on interchanging the positions of R_1 and R_2, if we get null point at l_2 distance then

$$\frac{R_2}{R_1} = \frac{\ell_2 + \alpha}{100 - \ell_2 + \beta} \quad \ldots(2)$$

Solving equation (1) and (2) get,

$$\alpha = \frac{R_2\ell_1 - R_1\ell_2}{R_1 - R_2} \text{ and } \beta = \frac{R_1\ell_1 - R_2\ell_2}{R_1 - R_2} - 100$$

These end corrections (α and β) are used to modify the observations.

Q. **In the meter bridge experiment, for R = 100 Ω and S = 200 Ω, no deflection is observed at a length of 33.0 cm. On interchanging the resistances, no deflection is observed at a length of 67.0 cm. Find the value of the end corrections α and β.**

Sol. $\alpha = \frac{R_2\ell_1 - R_1\ell_2}{R_1 - R_2} = \frac{(200)(33) - (100)(67)}{100 - 200} = 1\text{cm}$

$$\beta = \frac{R_1\ell_1 - R_2\ell_2}{R_1 - R_2} - 100 = \frac{(33)(100) - (200)(67)}{100 - 200} - 100 = 1\text{cm}$$

Maximum Permissible Error in ρ

The specific resistivity can be written as, $\rho = \frac{\pi D^2 S}{4L}\frac{\ell}{100-\ell}$

For no error in the value of S and the length of the wire L, maximum permissible error in ρ due to error in measurement of l (balance length) and D (diameter of wire),

$$\ln \rho = \ln\left(\frac{\pi S}{4L}\right) + 2\ln D + \ln l - \ln(100 - l)$$

Differentiating both sides, we get,

$$\frac{d\rho}{\rho} = 2\frac{dD}{D} + \frac{dl}{l} - \frac{d(100-l)}{(100-l)} = 2\frac{dD}{D} + \frac{dl}{l} + \frac{dl}{100-l}$$

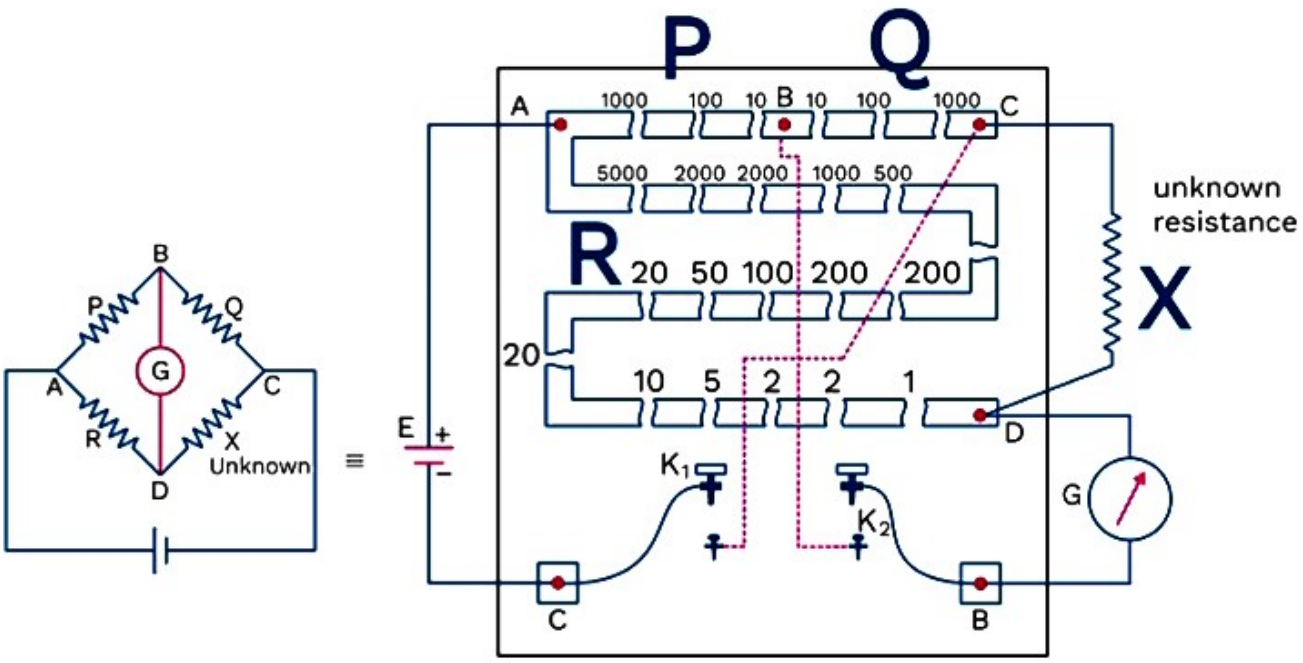
$$\left(\frac{\Delta\rho}{\rho}\right)_{max} = 2\frac{\Delta D}{D} + \frac{\Delta l}{l} + \frac{\Delta l}{100-l}$$

$\left(\frac{\Delta\rho}{\rho}\right)_{max}$ due to error in ℓ only is $= \frac{\Delta l}{l} + \frac{\Delta l}{100-l} = \frac{\Delta l(100)}{l(100-l)}$

$\left(\frac{\Delta\rho}{\rho}\right)_{max}$ will be least when l(100 – l) is maximum, i.e., l = 50 cm

So % error in resistance (resistivity) will be minimum if the balance point is at the mid-point of meter bridge wire.

9 Post Office Box



In a Wheatstone-Bridge circuit, if if $\frac{P}{Q} = \frac{R}{X}$ then the bridge is balanced. So unknown resistance $X = \frac{P}{Q}R = \frac{R}{(P/Q)}$.

A post office box is similar to a Wheatstone-Bridge circuit.

In the arms AB and BC, we can get any ratio of $\frac{P}{Q}$ by putting 10 Ω, 100 Ω or 1000 Ω resistances in the slots.

These arms are called ratio arms. Initially P = 10 Ω and Q = 10 Ω are taken to set $\frac{P}{Q} = 1$.

A battery is connected across points A and C and the unknown resistance X is connected across C and D.

Now, we start adding resistance in steps of 1 Ω in parts A to D and check the deflection in Galvanometer by first pressing key K_1 key then Galvanometer key K_2.

Suppose at R = 4 Ω, we get deflection toward left and at R = 5 Ω, we get deflection toward right. Then it can be said that the bridge will be balanced for a value of R lying between 4 and 5.

Now $X = \frac{R}{(P/Q)} = \frac{R}{(10/10)}$, $R \in (4, 5)$

So, we can estimate that X should be between 4Ω and 5Ω.

To get closer X, in the second observation, lets choose $\frac{P}{Q} = 10$ i.e. $\left(\frac{P = 100}{Q = 10}\right)$.

Suppose now at R = 42, we are getting deflection toward left, and at R = 43, deflection is toward right. So, $R \in (42, 43)$.

Now $X = \frac{R}{(P/Q)} = \frac{R}{(100/10)} = \frac{1}{10} R$ where $R \in (42, 43)$.

So, we can estimate that X lies between 4.2 and 4.3.

Now to get further closer, choose $\frac{P}{Q} = 100$. As we increase the $\frac{P}{Q}$ ratio, R will be divided by a greater number, so the answer will be up to more decimal places and will be more accurate.

Q. **In a post office box experiment, consider the ratio of $\frac{Q}{P} = \frac{1}{10}$. For a value of R = 192 Ω, the deflection in galvanometer is towards right, for a value of R = 193 Ω, the deflection in galvanometer is towards right but for a value of R = 194 Ω, the deflection in galvanometer is towards left. Then, find the range in which the unknown resistance lies.**

Sol. Deflection of galvanometer changes when R changes from 193Ω to 194Ω.

$X = \frac{Q}{R}(R) = \frac{1}{10}(193 \leftrightarrow 194) = 19.3 \leftrightarrow 19.4$

Unknown resistance lies between 19.3Ω and 19.4Ω.